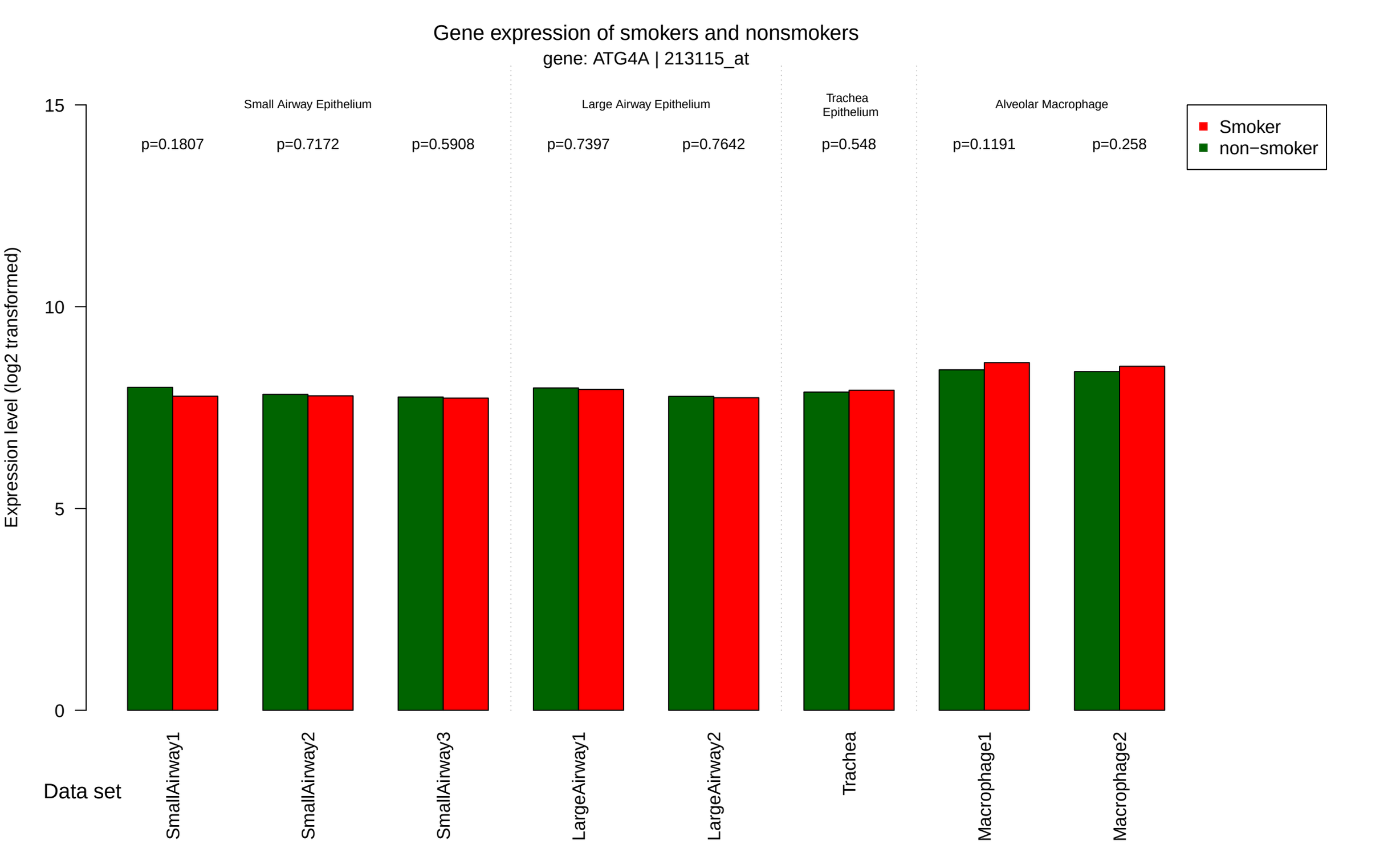
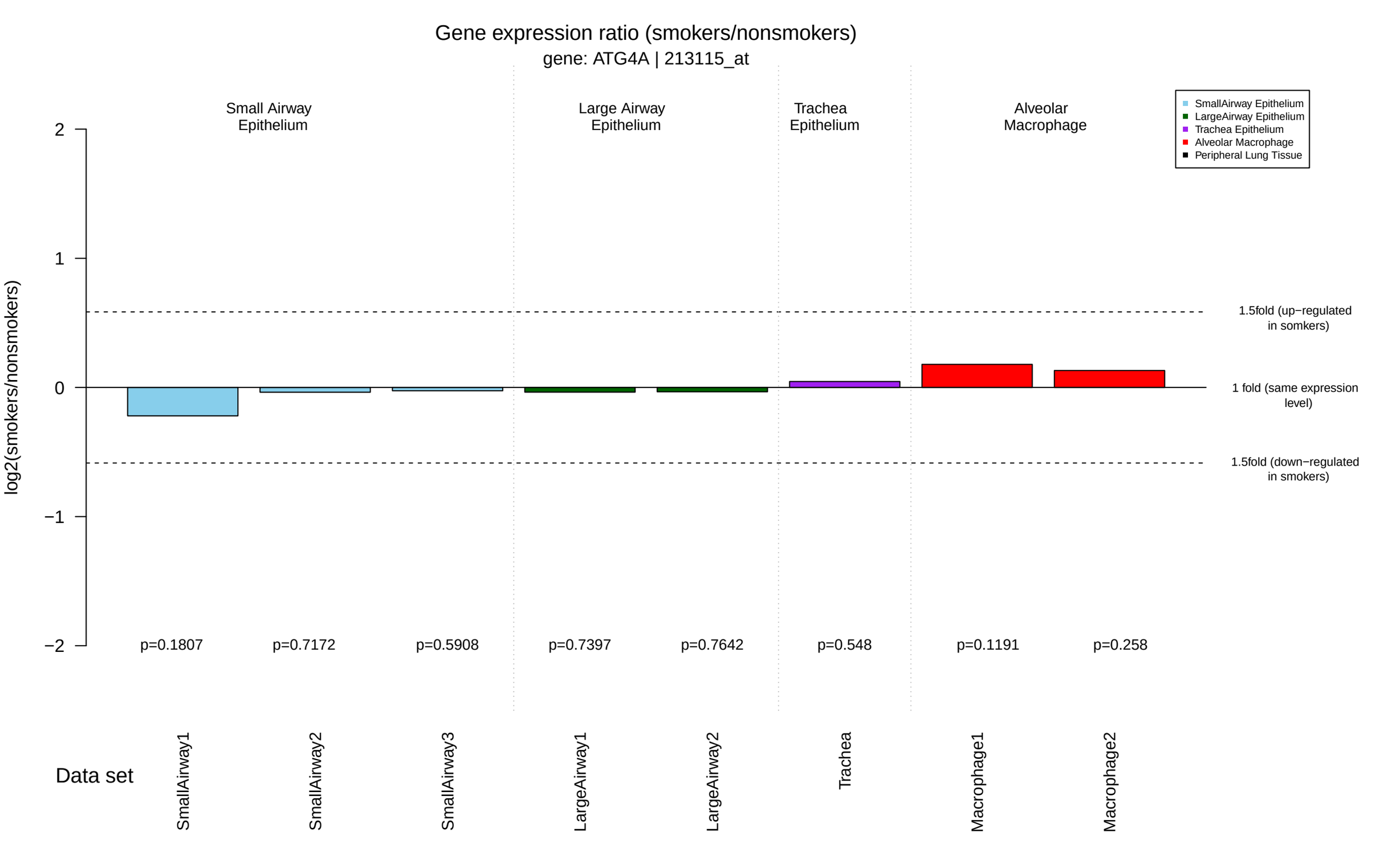
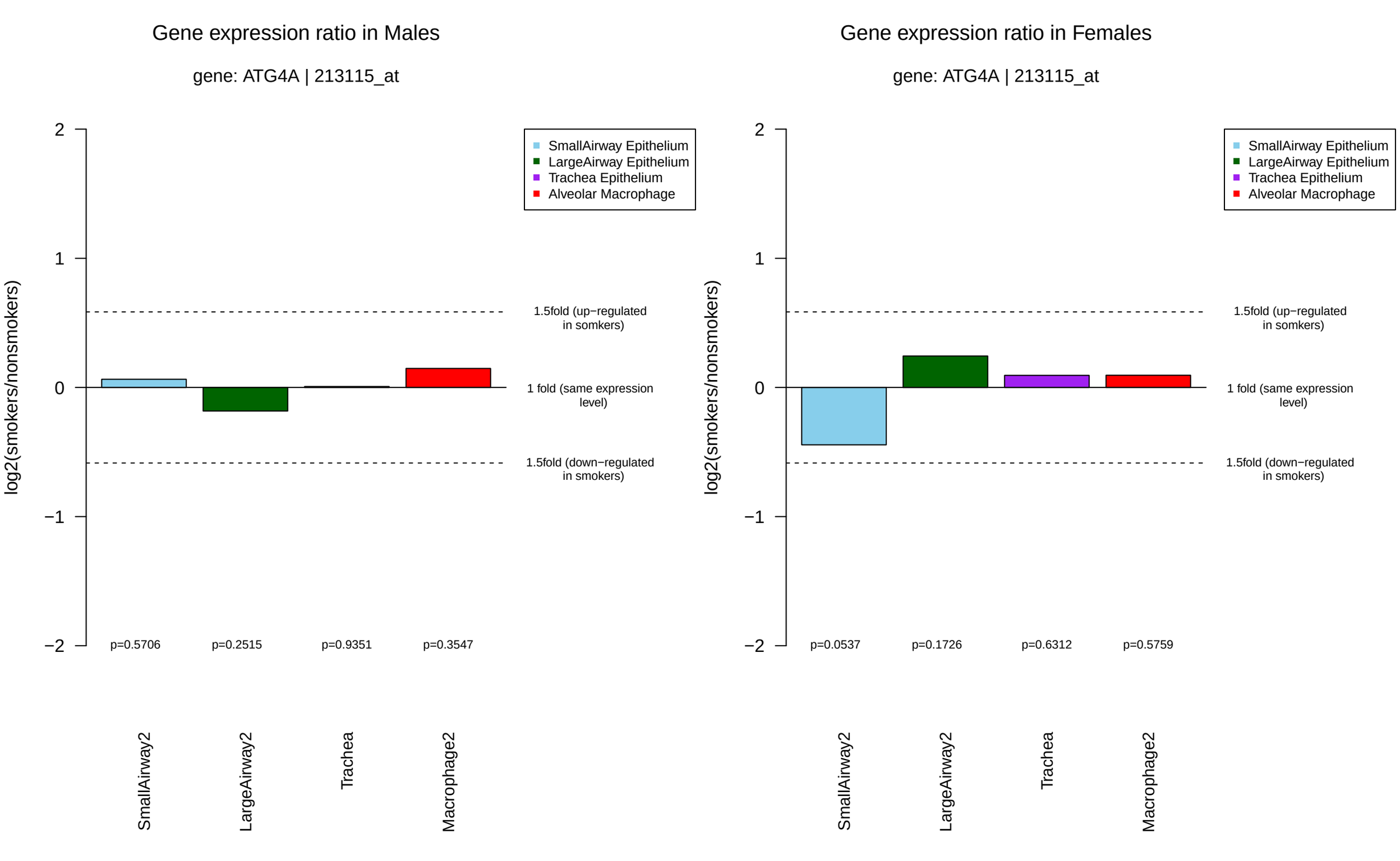
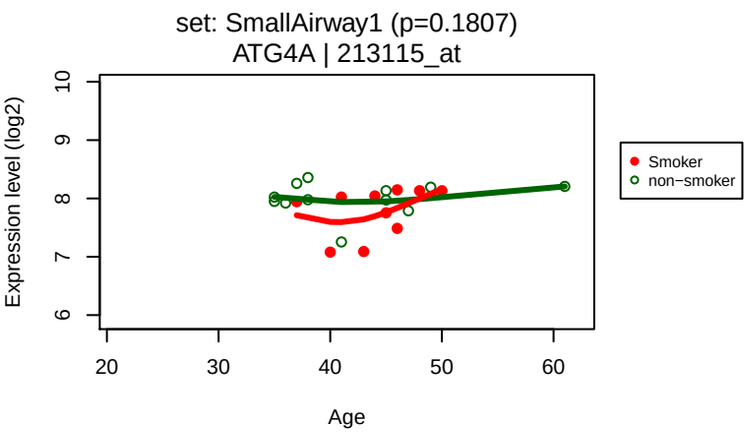
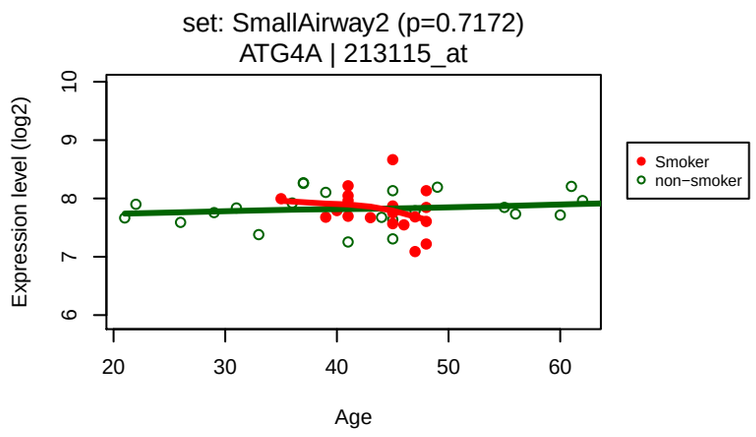
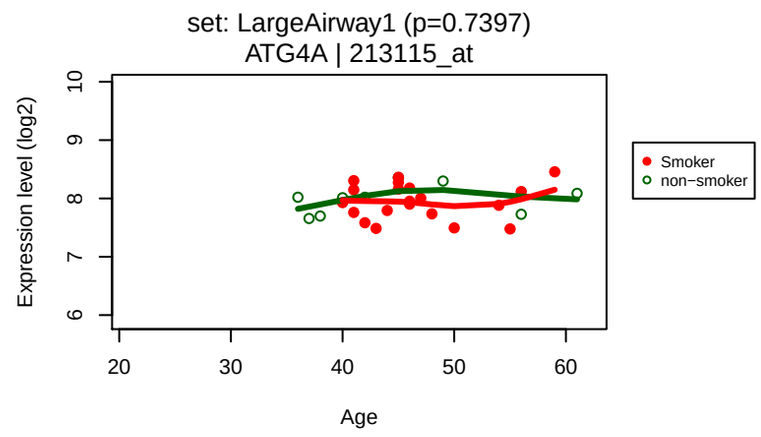

Gene expression of smokers and nonsmokers
gene: ATG4A | 213115_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1807
p=0.7172
p=0.5908
p=0.7397
p=0.7642
p=0.548
p=0.1191
p=0.258
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

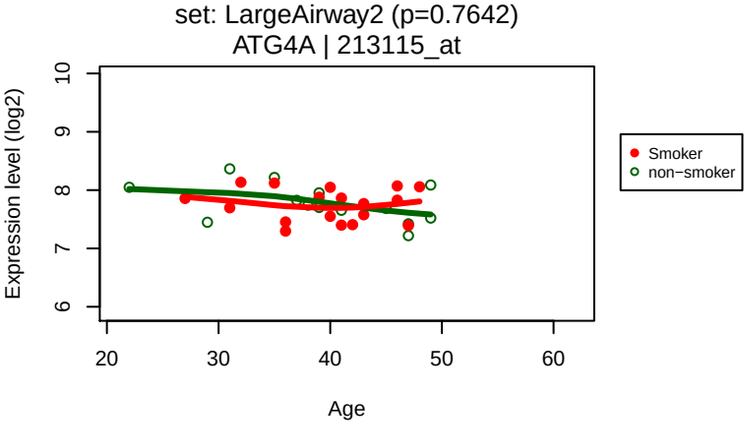

Gene expression ratio (smokers/nonsmokers)
gene: ATG4A | 213115_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1807
p=0.7172
p=0.5908
p=0.7397
p=0.7642
p=0.548
p=0.1191
p=0.258
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

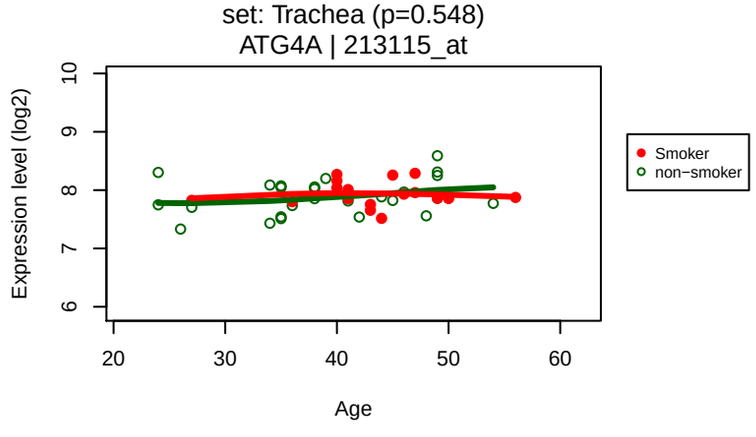

Gene expression ratio in Males
gene: ATG4A | 213115_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5706
p=0.2515
p=0.9351
p=0.3547
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATG4A | 213115_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0537
p=0.1726
p=0.6312
p=0.5759
SmallAirway2
LargeAirway2
Trachea
Macrophage2

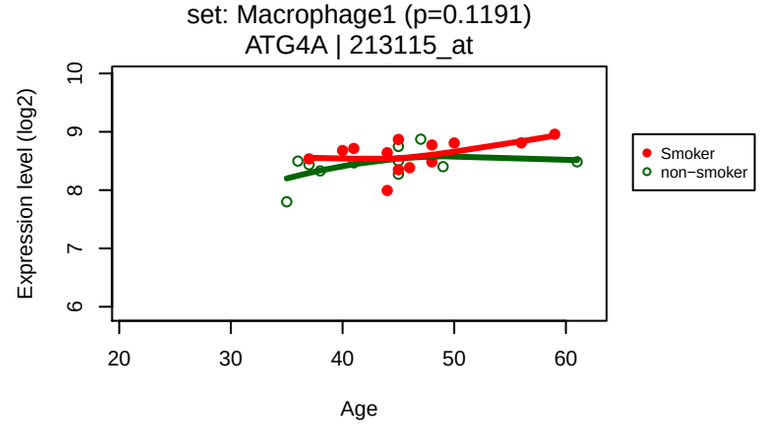

set: SmallAirway1 (p=0.1807)
ATG4A | 213115_at
Expression level (log2)
Age
Smoker
non−smoker

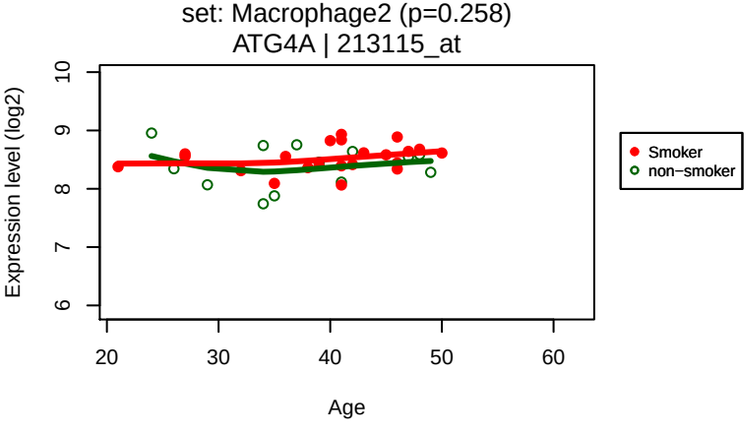

set: SmallAirway2 (p=0.7172)
ATG4A | 213115_at
Expression level (log2)
Age
Smoker
non−smoker

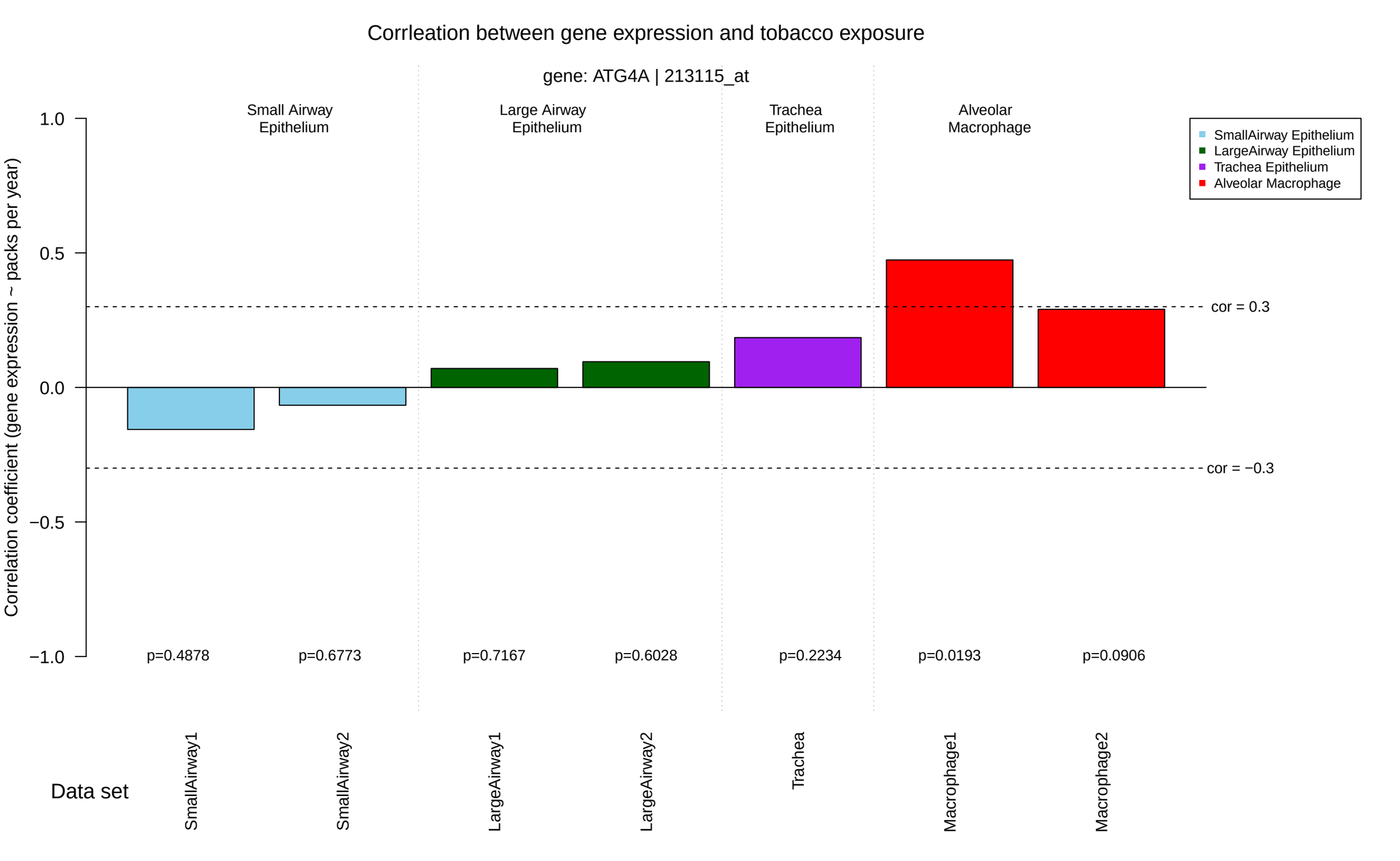

set: LargeAirway1 (p=0.7397)
ATG4A | 213115_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7642)
ATG4A | 213115_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.548)
ATG4A | 213115_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1191)
ATG4A | 213115_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.258)
ATG4A | 213115_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATG4A | 213115_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4878
p=0.6773
p=0.7167
p=0.6028
p=0.2234
p=0.0193
p=0.0906
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2